

CYBER SECURITY INCIDENT RESPONSE AND DIGITAL FORENSICS INVESTIGATION

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ABSTRACT

Cybersecurity incidents have become increasingly common due to the rapid growth of digital technologies. Organizations must respond quickly and effectively to minimize financial losses, protect sensitive information, and maintain business continuity.

This project presents a simulated Cyber Security Incident Response and Digital Forensics Investigation for a small IT organization. It demonstrates the complete incident handling process, including incident identification, evidence collection, analysis, containment, recovery, and recommendations.

The project also explains how digital forensic techniques can help identify the root cause of an incident and improve future cybersecurity preparedness.

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CHAPTER 1

1.1 Introduction

Cybersecurity incidents have become one of the biggest challenges for modern organizations. Cyber attacks such as phishing, ransomware, malware, and data breaches can disrupt business operations, compromise sensitive information, and cause significant financial losses.

Incident Response is a structured process used to detect, analyze, contain, eradicate, and recover from cybersecurity incidents. An effective incident response process minimizes the impact of attacks and helps organizations resume normal operations quickly.

Digital Forensics is the process of identifying, collecting, preserving, analyzing, and reporting digital evidence after a cybersecurity incident. It helps investigators determine how an attack occurred, what systems were affected, and what actions should be taken to prevent similar incidents in the future.

This project presents a simulated ransomware incident in a small IT organization. It explains the complete incident response lifecycle, evidence collection process, forensic findings, root cause analysis, and security recommendations based on industry best practices.

1.2 Objectives

- Understand the incident response lifecycle.
- Identify the source of a cybersecurity incident.
- Collect and analyze digital evidence.
- Perform root cause analysis.
- Recommend security improvements.
- Improve organizational cyber resilience.

1.3 Scope

This project focuses on a simulated ransomware incident affecting a small IT organization. It includes incident detection, evidence collection, forensic investigation, recovery activities, and security recommendations. No real systems were attacked or accessed during this project.

CHAPTER 2

INCIDENT OVERVIEW

2.1 Incident Scenario

ABC Technologies Pvt. Ltd. is a small IT organization with approximately 50 employees. On 15 June 2026, an employee received a phishing email that appeared to be from a trusted client. The email contained a malicious attachment named Invoice.pdf.exe.

The employee downloaded and executed the file. Within a few minutes, the ransomware spread across the employee's computer and encrypted several important files. The malware also attempted to access the company's shared network folder.

The IT team detected unusual file encryption activities and multiple failed login attempts. The affected system was immediately isolated from the network to prevent further damage.

2.2 Incident Details

Date- 15 June 2026

Affected Department- Finance

Affected System- Employee Laptop

Impact- File Encryption and Temporary
Service Disruption

Severity- High

2.3 Impact of the Incident

The incident resulted in the following impacts:

- .Temporary loss of access to important files.
- .Disruption of daily business operations.
- .Risk of confidential data exposure.
- .Financial loss due to system recovery.
- .Increased workload for the IT and Security teams.

The quick response by the IT team helped contain the attack before it spread to all organizational systems.

CHAPTER 3

EVIDENCE COLLECTION

3.1 Introduction

Evidence collection is a critical step in digital forensics. It involves identifying, preserving, and documenting digital evidence without altering its integrity. The collected evidence helps investigators understand how the incident occurred and supports the investigation process.

3.2 Evidence Collected

.E-001- Suspicious Email with Malicious Attachment::Employee Mailbox

.E-002- Employee Laptop System Logs
::Windows Event Logs

.E-003- Antivirus Detection Report
::Endpoint Security Software

.E-004- Firewall Logs
::Network Firewall

.E-005- Encrypted Files
::Employee Laptop

3.3 Evidence Preservation

To maintain the integrity of the investigation, the following actions were taken:

- .The affected laptop was isolated from the network.
- .System logs were collected and stored securely.
- .The suspicious email was preserved for analysis.
- .Evidence was documented with date and time.
- .Access to the evidence was restricted to the investigation team.

3.4 Findings

The collected evidence indicated that the ransomware infection originated from a phishing email containing a malicious attachment. The malware executed after the user opened the file, leading to file encryption on the affected system. Early detection and system isolation helped prevent the ransomware from spreading across the organization.

CHAPTER 4

INCIDENT RESPONSE PROCESS

4.1 Introduction

Incident Response is a structured approach used to detect, contain, investigate, recover from, and prevent cybersecurity incidents. A well-defined incident response process minimizes damage and helps restore normal business operations quickly.

4.2 Incident Response Lifecycle

1. Preparation

The organization maintained antivirus software, firewalls, system backups, and an incident response plan. Employees received basic cybersecurity awareness training.

2. Identification

The IT team detected unusual file encryption, multiple failed login attempts, and suspicious network activity. The incident was confirmed as a ransomware attack.

3. Containment

The affected laptop was immediately disconnected from the network to stop the ransomware from spreading. Access to shared folders was temporarily restricted.

4. Eradication

The malicious files were removed, the infected system was scanned using antivirus software, and all security patches were applied.

5. Recovery

The system was restored using clean backup files. All applications and services were tested before reconnecting the device to the network.

6. Lessons Learned

The organization reviewed the incident and implemented stronger security measures, including Multi-Factor Authentication (MFA), employee awareness training, and improved email filtering.

4.3 Outcome

The incident was successfully contained without affecting the entire network. The investigation helped identify the root cause and highlighted areas for improving the organization's cybersecurity posture.

CHAPTER 5

ROOT CAUSE ANALYSIS & DIGITAL FORENSICS FINDINGS

5.1 Root Cause Analysis

The investigation revealed that the ransomware attack was initiated through a phishing email containing a malicious attachment. The employee mistakenly opened the file without verifying its authenticity, allowing the malware to execute on the system.

The following factors contributed to the incident:

- .Lack of employee awareness about phishing attacks.
- .Absence of Multi-Factor Authentication (MFA).
- .Delayed software updates on the affected system.
- .Inadequate email filtering to block malicious attachments.

5.2 Digital Forensics Findings

The forensic investigation identified the following observations:

- .Email Analysis- Malicious attachment received through phishing email
- .System Logs- Unauthorized execution of a suspicious file.
- .Antivirus Logs- Ransomware detected after file execution.
- .Network Logs- Temporary communication with a suspicious external IP address.
- .File Analysis- Multiple files encrypted by ransomware.

5.3 Timeline of the Incident

- .09:10 AM- Phishing email received
- .09:15 AM- Malicious attachment opened
- .09:17 AM- Ransomware executed
- .09:20 AM- Files started encrypting
- .09:25 AM- IT team detected unusual activity
- .09:30 AM- System isolated from the network
- .11:00 AM- Recovery process started

5.4 Summary

The forensic investigation confirmed that the ransomware attack originated from a phishing email. Quick detection, system isolation, and backup recovery minimized the impact of the attack and prevented further damage to the organization's network.

CHAPTER 6

SECURITY RECOMMENDATIONS, CONCLUSION AND REFERENCES

6.1 Security Recommendations

Based on the investigation, the following recommendations are suggested to prevent similar cybersecurity incidents:

- .Enable Multi-Factor Authentication (MFA) for all user accounts.
- .Conduct regular cybersecurity awareness and phishing training for employees.
- .Keep operating systems and software up to date with the latest security patches.
- .Install and maintain updated antivirus and endpoint protection solutions.
- .Perform regular data backups and test recovery procedures.
- .Implement advanced email filtering to block malicious attachments.
- .Apply the Principle of Least Privilege (PoLP) to limit user access.
- .Monitor system logs and network activity for suspicious behavior.
- .Review and update the Incident Response Plan periodically.

6.2 Conclusion

This project demonstrated a simulated ransomware incident and the complete incident response process followed by a digital forensics investigation. The investigation identified the root cause of the attack, analyzed digital evidence, and recommended practical security improvements.

The project highlights the importance of timely incident detection, evidence preservation, employee awareness, and effective security controls. By implementing these recommendations, organizations can improve their cyber resilience and reduce the impact of future cyber attacks.

6.3 References

- 1.NIST Cybersecurity Framework (CSF).
- 2.CISA – Cybersecurity Best Practices.
- 3.OWASP Foundation – Security Guidelines.
- 4.Microsoft Security Documentation.
- 5.Cisco Cybersecurity Resources.